

1.0 Approach to the Otolaryngology–Head and Neck Surgery Patient

This book is organized into brief chapters addressing specific clinical entities. To enable readers to focus readily on their information needs, the chapters are arranged in a similar manner:

- Key Features
- Epidemiology
- Clinical
 - Signs and symptoms
 - Differential diagnosis
- Evaluation, including history, exam, imaging, and other testing
- Treatment options, including medical and surgical treatments
- Follow-up care

This first chapter is an exception because it deals entirely with the evaluation step. Specifically, we review in detail the approach to an efficient and effective otolaryngology patient history and physical examination, which should be especially useful to those new to the care of such patients.

◆ History

The generally accepted organization of the history and physical examination for a new patient is outlined in **Table 1.1**.

The History of Present Illness is the subjective narrative regarding the current problem. It should include a focused summary of the complaint, including location, time of onset, course, quality, severity, duration, associated problems, and previous testing or treatment.

◆ Physical Exam

The physical examination in otolaryngology is typically a complete head and neck exam. This should include an evaluation of the following:

General

- The general appearance of the patient (i.e., well- or ill-appearing, acute distress)
- Vital signs (temperature, heart rate, blood pressure, respiratory rate, weight, possibly BMI)
- Stridor, abnormal respiratory effort/increased work of breathing

Head

- Normocephalic, evidence of trauma
- Description of any cutaneous lesions of the head and neck

Ear

- Pinnae, ear canals, tympanic membranes, including mobility
- 512-Hz tuning fork testing (Weber, Rinne)

Nose

- External nasal deformities
- Anterior rhinoscopy noting edema, masses, mucus, purulence, septal deviation, perforation

Oral Cavity/Oropharynx

- Noting any masses, mucosal lesions, asymmetries, condition of dentition, presence/absence of tonsils and appearance
- Consider palpation of floor of mouth and base of tongue
- Hypopharynx and larynx
- Presence of hoarseness or phonatory abnormality
- Direct fiber optic or indirect mirror exam of the nasopharynx, hypopharynx, and larynx
- Laryngeal exam should note vocal fold mobility, mucosal lesions, and masses as well as assess the base of the tongue, valleculae, epiglottis, vocal folds, and piriform sinuses

Neck

- Inspection and palpation of the parotid and submandibular glands
- Inspection and palpation of the neck for adenopathy or masses
- Inspection and palpation of the thyroid gland for enlargement or masses
- Cranial nerve function

Other, more specialized aspects of an examination are discussed in the various sections that follow, such as vertigo assessment and nasal endoscopy.

◆ Endoscopic Exam

If the mirror examination does not provide an adequate assessment of the nasopharynx, hypopharynx, or larynx, a flexible fiberoptic nasolaryngoscopy is performed. Usually, the nose is decongested with oxymetazoline (Afrin, Schering-Plough Healthcare Products Inc., Memphis, TN) or phenylephrine (Neo-Synephrine, Bayer Consumer Health, Morristown, NJ) spray. Topical Pontocaine or lidocaine spray may be added for anesthetic. Surgilube jelly (E. Fougera & Co., Melville, NY) is helpful to reduce irritation. Antifog is applied to the tip of the flexible laryngoscope. The patient is best examined sitting upright. The tip of the scope is inserted into the nostril and under direct vision is advanced inferiorly along the floor of the nose into the nasopharynx. If septal spurring or other intranasal deformities prevent

advancement of the scope, the other nostril may be used. The nasopharynx is assessed for masses or asymmetry, adenoid hypertrophy, and infection. In the sleep apnea patient, the presence of anteroposterior (AP) or lateral collapse of the retropalatal region is remarked. The scope is then guided inferiorly to examine the base of the tongue, valleculae, epiglottis, piriform sinuses (piriform fossae), arytenoids, and vocal folds. Again, mucosal lesions, masses, asymmetries, and vocal fold mobility are noted. Asking the patient to cough, sniff, and phonate will reveal vocal fold motion abnormalities. The piriform sinuses may be better visualized if the patient puffs out the cheeks (exhalation with closed lips and palate).

◆ Other Tests

Often, laboratory studies, audiograms, or imaging studies are reviewed. These are summarized in the note after the physical exam section. Whenever possible, radiology images (CT, MRI) should be personally reviewed to confirm that one agrees with the reports.

◆ Impression and Plan

In the documentation of the patient’s visit, the note concludes with an impression and plan. Generally, a concise differential diagnosis is given, listing the entities that are considered most relevant. A plan is then discussed, including further tests to confirm or exclude possible diagnoses as well as medical or surgical treatments that will be instituted or considered. Timing of a return or follow-up visit, if needed, is noted.

A copy of one’s note, or a separate letter, should always be sent to referring physicians.

Table 1.1 Organization of the history and physical exam for a new patient

Chief complaint
History of present illness
Past medical history
Past surgical history
Current medications
Medication allergies
Social history
Family history
Review of systems
Physical examination
Laboratory testing/imaging
Impression
Plan

1.1 Diagnostic Imaging of the Head and Neck

Many of the structures of the head and neck are deep and inaccessible to direct visualization, palpation, or inspection. Therefore, valuable information may be obtained by the use of various radiographic techniques. Advances in technology have supplemented simple X-ray procedures with computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, and positron emission tomography (PET). Other imaging modalities are used for specific conditions, such as angiography for vascular lesions or barium swallow cinefluoroscopy for swallowing evaluations.

◆ Computed Tomography

A contrast-enhanced CT scan is typically the first imaging technique used to evaluate many ear, nose, throat, and head and neck pathologies. The CT scan is an excellent method for the staging of tumors and identifying lymphadenopathy. A high-resolution CT scan may be used in cases of trauma to the head, neck, laryngeal structures, facial bones, and temporal bone. Temporal bone CT is used to assess middle ear and mastoid disease; paranasal sinus CT is the gold standard test for assessing for the presence and extent of rhinosinusitis and many of its complications. A CT scan is superior to MRI in evaluating bony cortex erosion from tumor. A CT scan is also widely used for posttreatment surveillance of head and neck cancer patients.

Working Principle of CT

In CT, the X-ray tube revolves around the craniocaudal axis of the patient. A beam of X-rays passes through the body and hits a ring of detectors. The incoming radiation is continuously registered, and the signal is digitized and fed into a data matrix, taking into account the varying beam angulations. The data matrix can then be transformed into an output image (**Fig. 1.1**). The result is usually displayed in “slices” cross-sectionally. Different tissues attenuate radiation to varying degrees, allowing for the differentiation of tissue subtypes (**Table 1.2**). This absorption is measured in Hounsfield units. When one views an image, two values are displayed with the image: Window and Level. The Window refers to the range of Hounsfield units displayed across the spectrum from black (low) through the grayscale to white (high). Level refers to the Hounsfield unit on which middle gray is centered. By adjusting the window and level, certain features of the image can be better assessed or emphasized.

Recent advances have improved the quality of CT imaging. Multidetector scanners have several rows of photoreceptors, enabling the simultaneous acquisition of several slices. Helical techniques allow the patient table to move continuously through the scanner instead of stopping for each slice. These advances have significantly decreased scan times and radiation exposure while improving spatial resolution. Improved resolution and computing power enable cross-sectional images to be reformatted into any plane (axial, coronal, sagittal), as well as three-dimensional anatomy or subtraction images to be displayed when necessary or helpful (e.g., three-dimensional reconstruction of airways). Newer in-office flat-plate cone-beam scanners

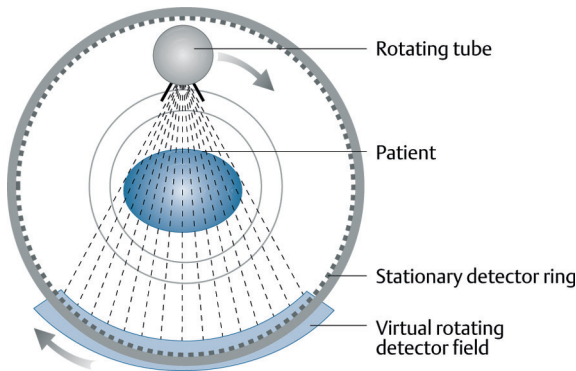


Fig. 1.1 Working principle of computed tomography. The X-ray tube revolves continuously around the longitudinal axis of the patient. A rotating curved detector field opposite to the tube registers the attenuated fan beam after it has passed through the patient. Taking into account the tube position at each time point of measurement, the resulting attenuation values are fed into a data matrix and further computed to create an image. (Used with permission from Eastman GW, Wald C, Crossin J. *Getting Started in Clinical Radiology: From Image to Diagnosis*. New York: Thieme; 2006:9.)

Table 1.2 Attenuation of different body components

Body Component	Hounsfield Units (HU)
Bone	1000–2000
Thrombus	60–100
Liver	50–70
Spleen	40–50
Kidney	25–45
White brain matter	20–35
Gray brain matter	35–45
Water	–5 to 5
Fat	–100 to –25
Lung	–1000 to –400

Data from Eastman GW, Wald C, Crossin J. *Getting Started in Clinical Radiology: From Image to Diagnosis*. Stuttgart/New York: Thieme; 2006.

can rapidly acquire 1-mm slice thickness images of the sinuses and temporal bone with very low radiation exposure.

Contrast Media

Intravenous contrast media are used in CT to visualize vessels and the vascularization of different organ systems. This allows better differentiation of vessels versus other structures. Some tissues also take up greater amounts of contrast natively, as well as in certain disease states (e.g., infection, neoplasm, edema). Luminal contrast material containing iodine or barium

can also be used in some structures (e.g., gastrointestinal tract) to clarify anatomy.

Computer-Assisted Surgical Navigation

CT scanning data can be utilized for computer-assisted surgical navigation. There are several systems in use. The axial CT image data, acquired at 1-mm slice thickness or less, are loaded onto the image guidance system in the operating room. The system utilizes the CT data and compares them to the patient’s facial features or landmarks via an infrared camera or electromagnetic field disturbance to determine point location in three-dimensional space. Various surgical instruments can be registered and detected. The location of an instrument tip is then displayed on the previous CT images in three planes. This is most often used in sinus and skull base surgery.

♦ Magnetic Resonance Imaging

MRI provides the physician with high-definition imaging of soft tissue without exposing the patient to ionizing radiation. MRI is useful for detecting mucosal tumors, neoplastic invasion of bone marrow, and, at times, perineural invasion of large nerves. MRI is valuable in assessing intracranial extension of tumors of the head and neck. Gadolinium-enhanced MRI of the brain with attention to the internal auditory canal is the gold standard test for diagnosis of vestibular schwannoma or meningioma, easily identifiable on postcontrast T1-weighted images. The disadvantages of MRI include limited definition of bony detail and cost. Magnetic resonance angiography (MRA) is a useful modality for imaging vascular anatomy or vascular pathology without the intravascular infusion of iodine contrast medium, which is used in traditional angiography with X-ray fluoroscopy.

Working Principle of MRI

MRI is a technique that produces cross-sectional images in any plane without the use of ionizing radiation. MR images are obtained by the interaction of hydrogen nuclei (protons), high magnetic fields, and radiofrequency pulses. This is done by placing the patient in a strong magnetic field, which initially aligns the hydrogen nuclei in similar directions. The intensity of the MRI signal that is converted to imaging data depends on the density of the hydrogen nuclei in the examined tissue (i.e., mucosa, fat, bone) and on two magnetic relaxation times (Table 1.3). MRI imaging is more time-intensive to perform and thus more vulnerable to motion

Table 1.3 Definitions of terms used in magnetic resonance imaging

TR	Time to repetition
TE	Time to echo
Ti	Time to inversion
T1	Time to magnetize (regrowth); also known as spin lattice relaxation time
T2	Time to demagnetize (DK); also known as spin relaxation time

artifact, as the patient must remain still for minutes (versus seconds in CT imaging).

Contraindications to MRI

- Implanted neural stimulators, cochlear implants, and cardiac pacemakers (MRI may cause temporary or permanent malfunction)
- Ferromagnetic aneurysm clips or other foreign bodies with a large component of iron or cobalt, which may move within the body or heat up in the MRI scanner
- Metallic fragments within the eye (e.g., may be seen in patients who weld or grind metals)
- Placement of a vascular stent, coil, or filter in the past 6 weeks
- Ferromagnetic shrapnel
- Relative contraindications include claustrophobic patients, critically ill patients, morbidly obese patients who cannot physically fit in the MRI scanner, and those having metal implants in the region of interest and possibly tattoos with ferromagnetic ink.

◆ Ultrasound

Ultrasound or ultrasonography is an inexpensive and safe method of gaining real-time images of structures of the head and neck. Neck masses can be assessed for size, morphologic character (i.e., solid, cystic, or combined solid and cystic, also known as complex), and association with adjacent structures. Vascularity may also be assessed. High-resolution ultrasound is used for head and neck anomalies such as thyroglossal duct cysts, branchial cleft cyst, cystic hygromas, salivary gland masses, abscesses, carotid body and vascular tumors, and thyroid masses. There is growing emphasis on using ultrasound, when possible, in pediatric patients to reduce the use of ionizing radiation from CT imaging.

Ultrasound, combined with fine-needle aspiration biopsy (FNAB) and cytology, is helpful both in providing a visual description and as an aid for specific localization sampling of a mass for cytologic evaluation. Until recently, ultrasounds were performed mainly by radiologists. However, many otolaryngologists are now performing their own in-office ultrasounds and ultrasound-guided FNABs.

Working Principle of Ultrasound

An alternating electric current is sent through a piezoelectric crystal; the crystal vibrates with the frequency of the current, producing sound waves of that frequency. In medical ultrasound, typical frequencies vary between 1 and 15 MHz. Ultrasound gel acoustically couples the ultrasound transducer to the body, where the ultrasound waves can then spread. Inside the body the sound is absorbed, scattered, or reflected. Fluid-filled (cystic) structures appear dark and show acoustic enhancement behind them. Bone and air appear bright because they absorb and reflect the sound, showing an “acoustic shadow” behind them (**Fig. 1.2**). Linear transducers with a width

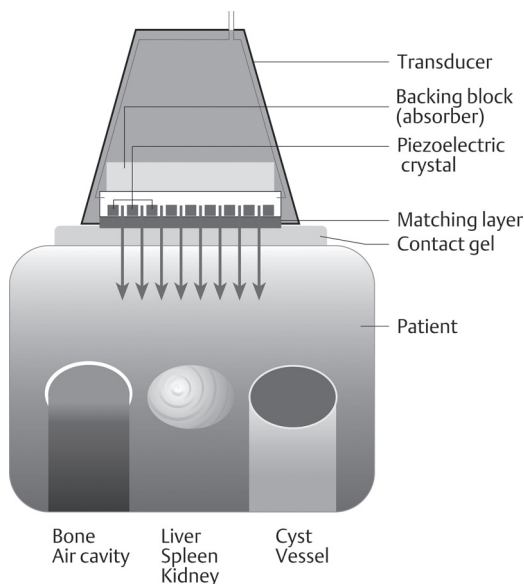


Fig. 1.2 Working principle of ultrasonography. (Used with permission from Eastman GW, Wald C, Crossin J. *Getting Started in Clinical Radiology: From Image to Diagnosis*. New York: Thieme; 2006:11.)

of 7.5 to 9 cm and frequencies of 10 to 13 MHz are typically used for evaluating the neck and thyroid.

◆ Barium Esophagram

An esophagram (also known as a barium swallow) is designed to evaluate the pharyngeal and esophageal mucosa; it is distinct from a modified barium swallow (MBS), which evaluates functional aspects of the upper swallow process and is usually performed in conjunction with a speech pathologist. These techniques are performed utilizing fluoroscopy. Fluoroscopy with intraluminal contrast is invaluable for studying the functional dynamics of the pharynx and esophagus.

An MBS evaluates the coordination of the swallow reflex. It is most often used to determine the etiology and severity of food bolus processing issues as well as the risk and/or presence of airway aspiration. A speech pathologist is usually in attendance and administers barium suspensions of varying thickness (thin liquid, thick liquid, nectar, paste, and solid) while the radiologist observes fluoroscopically, primarily in the lateral but also in the anteroposterior projections. One can also assess for esophageal motility/dysmotility, Zenker diverticulum, stricture, mass, hiatal hernia, or obvious free reflux.

Contrast Media

Barium suspension is the most commonly used fluoroscopic contrast agent. If a perforation of the hypopharynx or esophagus is suspected, there is a risk for barium extravasation into the soft tissues of the neck or chest. Therefore, in these cases, water-soluble contrast agents are often used (such as Gastrografin, Bracco Diagnostics, Inc., Princeton, NJ). However, it is important to note that these agents may cause a chemical pneumonitis or severe pulmonary edema if aspirated into the airway, so they must be cautiously used if concern for aspiration exists.

◆ Nuclear Medicine Imaging

Positron Emission Tomography with Computed Tomography

PET-CT is essentially a positron emission tomography scan performed and superimposed upon a simultaneous computed tomography scan to allow precise correlation between increased function (enhanced cellular activity) and anatomic evaluation provided by the CT. The PET scanning portion is a functional imaging technique that measures metabolic activity through the use of molecules tagged with positron-emitting isotopes such as the glucose precursor ^{18}F -fluoro-deoxyglucose (FDG), the most commonly used radiotracer, which has a half-life of ~ 110 minutes. ^{18}F is produced in a cyclotron and immediately extracted from solution, incorporated into the carbohydrate molecule, and administered.

Working Principle of PET-CT

After being emitted from the atom, the positron travels in the tissue for a short distance until it encounters an electron and forms a positronium, which immediately annihilates (converts its mass to energy), forming two photons. These annihilation photons travel in opposite directions from each other and are picked up by the detectors placed around the patient. Simultaneous detection of these photons relates them to the same annihilation event, allowing the event to be localized in space. Detection of annihilation by the dedicated PET scanner yields spatial resolution and sensitivity. The spatial resolution of the final reconstructed images is limited by the number of collected events.

FDG is taken up by glucose transporters. Normally, glucose enters into a cell, is phosphorylated by hexokinase, and then enters directly into either the glycolytic or glycogenic pathway. FDG, a glucose analogue, is subsequently unable to continue into the usual glucose metabolic pathways (because of the substitution of fluorine for a hydroxyl group) and is essentially trapped in the cell as FDG-phosphate. Because neoplastic cells have higher rates of glycolysis and glucose uptake, localized areas of intracellular activity on a PET scan may represent neoplastic disease. Tumor concentration of FDG generally peaks at 30 minutes, remains constant for 60 minutes, and then declines.

Note that FDG can also accumulate nonspecifically in other cells that have active glycolysis, such as organs with high glucose metabolism (e.g., brain and kidneys) as well as areas of active inflammation and infection. This may

lead to a false positive PET-CT scan. Other activities that may cause false positive findings include muscular activity, foreign bodies, and granulomas. False negatives in PET-CT scans may occur when the tumor threshold is too small (< 0.5 cm in diameter). In PET scanning, quantification of FDG uptake intensity is generally expressed on an arbitrary scale as standard uptake values (SUVs).

Thyroid Scintigraphy

Thyroid scintigraphy renders, at one point in time, information about the global and regional functional status of the thyroid. It is observer-independent and reproducible with low inherent radiation exposure. Scintigraphic imaging of the thyroid helps determine whether solitary or multiple nodules are functional when compared with the surrounding thyroid tissue. Findings for a nodule may be normal functional (warm), hyperfunctional (hot), or hypofunctional (cold). Scintigraphy can also help determine whether cervical masses contain thyroid tissue, and it can demonstrate whether metastases from well-differentiated thyroid cancer concentrate iodine for the purpose of radioiodine therapy. For thyroid scintigraphy the following radionuclides are in use: technetium-99m (^{99m}Tc), iodine-123 (^{123}I), and iodine-131 (^{131}I).

Working Principle of Thyroid Scintigraphy

The technique of thyroid scintigraphy is based on the principle that functional, active thyroid cells incorporate iodine and a gamma camera can then be used to detect the accumulated radionuclide.

Parathyroid Scintigraphy

Several radiotracers are available for parathyroid scintigraphy. At present, the radiotracer of choice is ^{99m}Tc -sestamibi (also called sestamibi, methoxy-isobutylisonitrile, or MIBI). ^{99m}Tc -sestamibi is a lipophilic cation that is taken up in the mitochondria of the cells. Of note, this radiotracer can be used with a wide variety of imaging techniques, including planar multiplex ion-beam imaging (MIBI), single-photon emission computed tomography (SPECT), and fused SPECT-CT.

Working Principle of Parathyroid Scintigraphy

Sestamibi accumulates in the thyroid and parathyroid tissues within minutes after IV administration, but it has a different washout rate from these two tissues. It is released faster from the thyroid than from the parathyroid. The presence of large numbers of mitochondria-rich cells in parathyroid adenomas is thought to be responsible for their slower release of ^{99m}Tc -sestamibi from hyperfunctioning parathyroid tissue than from the adjacent thyroid tissue. Thus, on 2- to 3-hour washout images, after thyroid uptake has dissipated, the presence of a retained area of activity allows one to identify and localize a parathyroid adenoma. Overall, ^{99m}Tc -sestamibi parathyroid scintigraphy has good sensitivity for the detection and localization of a single adenoma in patients with primary hyperparathyroidism. Correlation with ultrasound findings can also be helpful.

Single-Photon Emission Computed Tomography

SPECT scanning for parathyroid disease allows increased accuracy of routine sestamibi scanning by ~ 2 to 3%. SPECT scanning can be performed within the first several hours after a patient is injected with the sestamibi. During the scan, multiple images are taken of the patient's head and neck. These images are assembled to provide a three-dimensional picture. SPECT scanning is typically used when ordinary planar sestamibi scans are inconclusive.

Four-Dimensional Parathyroid CT (4D-CT)

4D-CT is a specialized imaging study that was first introduced in 2006 for the purpose of identifying abnormal parathyroid glands in the planning of parathyroid surgery. 4D-CT includes image sets in three planes (axial, coronal, sagittal) from the angle of the mandible to the mediastinum. The "fourth" dimension of 4D-CT is the perfusion information derived from multiple contrast phases. It is most commonly performed with three phases: noncontrast, arterial, and delayed-phase imaging. 4D-CT has a higher radiation dose than scintigraphy.

1.2 Hematology for the Otolaryngologist

◆ Key Features

- Head and neck surgery patients may require screening for, and correction of, hematologic disorders.

An overview of blood components, disorders, and transfusion complications is provided in this chapter.

◆ Blood Loss Management

Estimated Blood Volume (EBV)

- 95–100 mL/kg for premature infant
- 85–90 mL/kg for full-term infant
- 80 mL/kg for infants up to 12 months
- 70–75 mL/kg for adult males
- 65–70 mL/kg for adult females

Allowable blood loss = $[\text{EBV} \times (\text{Hct} - \text{target Hct})] / \text{Hct}$

Replace every 1 mL blood loss with 3 mL crystalloid, 1 mL colloid, or 1 mL packed red blood cells (PRBCs).

PBRC Transfusion Guidelines

- One unit PRBCs increases Hct ~ 3% and hemoglobin (Hb) ~ 1 g/dL in adults.
- 10 mL/kg PRBCs increases Hct ~ 10%

Compatibility Testing

- **Type specific:** ABO-Rh typing only; 99.80% compatible.
- **Type and screen:** ABO-Rh and screen; 99.94% compatible.
- **Type and crossmatch:** ABO-Rh screen, and crossmatch; 99.95% compatible. Crossmatching confirms ABO-Rh typing, detects antibodies to the other blood group systems, and detects antibodies in low blood titers.
- **Screening donor blood:** Hematocrit is determined; if normal, the blood is typed, screened for antibodies, and tested for hepatitis B, hepatitis C, syphilis, human immunodeficiency virus-1 (HIV-1), HIV-2, and human T cell lymphotropic viruses 1 and 2.

Blood Component Therapy

The archaic perioperative axiom of transfusing patients to maintain Hb of 10 and a hematocrit of 30 has fallen by the wayside. Although these are indeed safe guidelines for patients with coronary artery disease, transfusions are currently guided by hemodynamics, intraoperative blood loss, and laboratory values such as the arterial blood gas. Blood replacement in a complex surgical patient is part of goal-directed fluid therapy guided by the clinical response seen in continuously monitored flow parameters (see **Chapter 2.3**).

- **Whole blood:** 40% hematocrit; used primarily in hemorrhagic shock.
- **PRBCs:** Each unit has a volume of 250 to 300 mL with a hematocrit of 70 to 85%.
- **Platelets:** A normal platelet count is 150,000 to 400,000/mm³. Thrombocytopenia is defined as <150,000/mm³. Intraoperative bleeding increases with counts between 40,000 and 70,000/mm³, and spontaneous bleeding can occur with counts <20,000/mm³. During most surgeries, platelet transfusions are probably not needed unless the count is less than 50,000/mm³. One unit of platelets will increase platelet count 5000 to 10,000/mm³. The usual dose is 1 unit of platelets per 10 kg body weight. Platelets are stored at room temperature; ABO compatibility is not necessary.
- **Fresh frozen plasma (FFP):** Acute reversal of warfarin requires 5 to 8 mL/kg of FFP. ABO compatibility is mandatory. A 250 mL bag contains all coagulation factors except platelets. 10 to 15 mL/kg will increase plasma coagulation factors to 30% of normal. Fibrinogen levels will increase by 1 mg/mL of plasma transfused.
- **Cryoprecipitate:** Indications include hypofibrinogenemia, von Willebrand's disease, and hemophilia A. ABO compatibility is not necessary. Each 10 to 20 mL/bag contains 100 units of factor VIII-C, 100 units of von Willebrand factor (vWF), 60 units of factor XIII, and 250 mg of fibrinogen.

◆ Massive Transfusions

A massive transfusion is defined as the replacement of a patient's total blood volume in less than 24 hours. It also applies to the acute administration of more than half the patient's estimated blood volume per hour.

◆ Universal Donor Blood

Group O, Rh-negative blood should be reserved for patients close to exsanguination. If time permits, crossmatched or uncrossmatched type-specific blood should be administered. Group O, Rh-negative blood should not be given as whole blood. The serum contains high anti-A and anti-B titers, which may cause hemolysis of recipient blood.

If more than 4 units of group O, Rh-negative whole blood is administered, type-specific blood should not be given subsequently because the potentially high anti-A and anti-B titers could cause hemolysis of the donor blood.

Patients administered up to 10 units of group O, Rh-negative PRBCs may be switched to type-specific blood, since there is an insignificant risk of hemolysis from the small volume of plasma administration with PRBCs.

◆ Complications of Transfusions

Immune Reactions (Hemolytic versus Nonhemolytic)

Hemolytic Reactions

- *Acute Hemolytic Reaction.* An acute hemolytic reaction occurs when ABO-incompatible blood is transfused, resulting in acute intravascular hemolysis; the severity of a reaction often depends on how much incompatible blood has been given. Symptoms include fever, chills, chest pain, anxiety, back pain, and dyspnea; in anesthetized patients, the reaction may present with fever, tachycardia, hypotension, hemoglobinuria, and diffuse oozing in the surgical field. Free Hb in plasma or urine is evidence of a hemolytic reaction. Risk of a fatal hemolytic transfusion reaction: 1:600,000 units.

- *Delayed Hemolytic Reaction.* Typically, this reaction is delayed 2 to 21 days after the transfusion. The symptoms are generally mild and may include malaise, jaundice, and fever; treatment is supportive. A delayed hemolytic reaction is caused by incompatibility of minor antigens (i.e., Kidd, Duffy, Kelly, etc.), causing hemolysis.

Nonhemolytic Reactions

- *Febrile Reaction.* A febrile reaction is the most common nonhemolytic reaction (0.5–1.0% of RBC transfusions and up to 30% of platelet transfusions). The reaction is the result of the action of recipient antibodies against donor antigens present on leukocytes and platelets; treatment includes stopping or slowing the infusion and antipyretics.

- *Urticarial Reaction.* An urticarial reaction occurs in 1% of transfusions; it is thought to be due to sensitization of the patient to transfused plasma proteins. It is characterized by erythema, hives, and itching without fever. Treat with antihistamine drugs.

- *Anaphylactic Reaction.* Anaphylactic reactions are rare (1:500,000). Patients with IgA deficiency may be at an increased risk because of the transfused IgA reaction with anti-IgA antibodies.

Transfusion-Related Lung Injury

Transfusion-related lung injury (TRALI) is due to a transfusion of anti-histocompatibility leukocyte antigen (HLA) antibodies that interact with and cause the patient's white cells to aggregate in the pulmonary circulation. Risk: 1:6000. Treatment is supportive, using a similar therapy as for managing acute respiratory distress syndrome (ARDS).

Graft-versus-Host Disease

Graft-versus-host disease is most commonly seen in immunocompromised patients. Cellular blood products contain lymphocytes capable of mounting an immune response against the compromised host.

Posttransfusion Purpura

Due to the development of platelet alloantibodies; the platelet count typically drops dramatically one week after the transfusion.

Immune Suppression

Transfusion of leukocyte-containing blood products appears to be immunosuppressive. Blood transfusions may increase the incidence of serious infections following surgery or trauma. Blood transfusions may worsen tumor recurrence and mortality rate following resections of many cancers.

Infectious Complications

Viral Infections

- **Hepatitis:** Risk of hepatitis B virus (HBV) is 1:137,000; risk of hepatitis C virus (HCV) is 1:1,000,000
- **HIV/acquired immunodeficiency syndrome (AIDS):** Risk of HIV is 1:1,900,000
- **Cytomegalovirus (CMV) and Epstein-Barr virus:** Common and usually cause asymptomatic or mild systemic illness

Bacterial Infections

Gram-positive and gram-negative bacteria contamination of blood is rare. Specific bacterial reactions transmitted by blood include syphilis, brucellosis, salmonellosis, yersiniosis, and various rickettsioses.

Parasitic Infections

Malaria, toxoplasmosis, and Chagas's disease are very rare.

◆ Other Complications Related to Blood Transfusion

Metabolic Abnormalities

- Decreased pH secondary to increased hydrogen ion production
- Increased potassium: Due to cell lysis (increases with length of storage)
- Decreased 2,3-diphosphoglycerate (2,3-DPG): Consumed by RBCs; P50 decreases to 18 mm Hg after 1 week and to 15 mm Hg after 3 weeks.

- Citrate toxicity: Citrate metabolism to bicarbonate may contribute to metabolic alkalosis; binding of calcium by citrate could result in hypocalcemia and the ability of the liver to metabolize citrate to bicarbonate.

Microaggregates

Microaggregates consisting of platelets and leukocytes form during the storage of whole blood. Micropore filters may help remove these particles.

Hypothermia

The use of blood warmers (except for platelets) greatly decreases the likelihood of transfusion-related hypothermia.

Coagulopathies

Coagulopathies occur after massive transfusions (>10 units).

Dilutional Thrombocytopenia

Dilutional thrombocytopenia is a common cause of abnormal bleeding in the setting of massive transfusion. It typically responds quickly to platelet transfusion.

Factor Depletion

Factors V and VIII are very labile in stored blood and may decrease to levels as low as 15 to 20% of normal (this is usually enough for hemostasis).

Disseminated Intravascular Coagulation

Disseminated intravascular coagulation (DIC) is a hypercoagulable state caused by activation of the clotting system leading to deposition of fibrin in microvasculature, causing secondary fibrinolysis, leading to consumption of platelets and factors.

◆ Treatment of Hemolytic Transfusion Reactions

1. Stop the transfusion.
2. Check for error in patient or donor identification.
3. Send donor unit and newly obtained blood sample to blood bank for re-crossmatch.
4. Treat hypotension with fluids and pressors as needed.
5. If transfusion is required, use type O-negative PRBCs and type AB FFP.
6. Maintain the urine output at a minimum of 75 to 100 mL/h by generously administering IV fluids; consider mannitol, 12.5 to 50 g, or furosemide, 20 to 40 mg.
7. Monitor for signs of DIC clinically or with laboratory tests.
8. Send patient blood sample for direct antiglobulin (Coombs) test, free Hb, haptoglobin; send urine for Hb.

◆ Coagulation Studies

Partial Thromboplastin Time (PTT)

- By adding particulate material to a blood sample, which activates the intrinsic coagulation system, the PTT can be determined.
- PTT measures the clotting ability of all factors in the intrinsic and common pathways except factor XIII.
- PTT is abnormal (prolonged) if there are decreased amounts of coagulation factors, if the patient is on heparin, or if there is a circulating anticoagulant present.
- Normal values are between 25 and 37 seconds.

Prothrombin Time (PT)

- Performed by measuring the time needed to form a clot when calcium and a tissue extract are added to plasma.
- PT evaluates the activity of fibrinogen, prothrombin, and factors V, VII, and X; evaluates the intrinsic coagulation system and vitamin K-dependent factors.
- Normal PT is 10 to 12 seconds (depending on control).

International Normalization Ratio (INR)

INR is a means of standardizing PT values: it is the ratio of the patient's PT to the control PT value. INR values are commonly used as a guide for warfarin therapy.

Activated Clotting Time (ACT)

The ACT is a simple test that provides a global measurement of hemostatic function and is measured after the whole blood is exposed to a specific activator of coagulation. The time for clot formation after whole blood is exposed to Celite (diatomaceous earth) is defined as the ACT.

There is a linear relationship between ACT and heparin dose, which provides a convenient method to measure response to heparin anticoagulation. Normal values are 90 to 120 seconds. The ACT lacks sensitivity to clotting abnormalities.

Platelet Function

Detailed evaluation of possible platelet aggregation disorders requires complex hematology testing. However, a screening platelet function test (via the use of the PFA analyzer) is available and can rapidly screen for platelet function abnormality without identifying a specific cause (e.g., recent aspirin use, von Willebrand's disease). Thus, abnormal PFA-100 test should prompt detailed work-up, whereas a normal test is reassuring in the setting of planned elective surgery, such as tonsillectomy.

◆ Hematologic Disorders

Sickle Cell Anemia

Sickle cell anemia is a hereditary hemolytic anemia resulting from the formation of an abnormal Hb (HbS); sickle Hb has less affinity for oxygen and decreased solubility. Patients are typically asymptomatic. Vigorous physical activity, high altitude, air travel in unpressurized planes, and anesthesia are potentially hazardous.

Clinical features include signs and symptoms of anemia (Hb levels 6.5–10 gm/dL), obstructive or hemolytic jaundice, lymphadenopathy, hematuria, epistaxis, priapism, finger clubbing, and skeletal deformities.

The disease manifests as periodic exaggeration of symptoms or sickle crises. There are four types of crises:

- Vaso-occlusive crisis: caused by sickled cells blocking the microvasculature; characterized by sudden pain frequently without a precipitating event.
- Hemolytic crisis: seen in patients with sickle cell disease and glucose-6-phosphate dehydrogenase (G6PD) deficiency; manifests as sudden hemolysis.
- Sequestration crisis: red blood cells are sequestered in the liver and spleen, leading to splenohepatomegaly and acute fall in hematocrit. This can progress to circulatory collapse.
- Aplastic crisis: characterized by transient episodes of bone marrow depression; often occurs after viral illness.

Management of Sickle Cell Anemia

The practice of transfusion to an end point of 70% HbA and less than 30% HbS preoperatively is controversial. Patients should be well oxygenated and hydrated intraoperatively to lessen the chance of sickling. Acidosis and hypothermia can also trigger a crisis and should also be avoided.

Factor Deficiencies

Factor VIII Deficiency (Hemophilia A)

The half-life of factor VIII is 8 to 12 hours. Treatment includes lyophilized factor VIII, cryoprecipitate, or desmopressin. Infusion of 1 unit of factor VIII per kg will increase the factor VIII activity by 2%. Levels of 20 to 40% are desirable prior to surgery.

Factor IX Deficiency (Hemophilia B; Christmas's disease)

The half-life of factor IX is 24 hours. Therapy consists of factor IX concentrates of FFP. To ensure surgical hemostasis, activity levels of 50 to 80% are needed. Infusion of 1 unit per kg of factor IX will increase the factor IX level by 1%.

von Willebrand's Disease (VWD)

VWD is the most common of all the inherited bleeding disorders. It occurs in ~1 of every 100 to 1000 people. VWD affects both males and females, with

three forms. von Willebrand factor (VWF) is involved in platelet adhesion and in carrying factor VIII.

- Type 1 VWD: Most common, usually mild; low VWF, possibly low factor VIII.
- Type 2 VWD: Abnormal function of VWF; various forms.
- Type 3 VWD: Rare, severe. Usually no VWF and low factor VIII.

Testing for VWD includes platelet function assay, VWF levels, ristocetin cofactor assay to assess VWF function, factor VIII levels, and VWF multimers.

Management of von Willebrand's Disease

The goal of therapy is to correct the defect in platelet adhesiveness (by raising the level of effective VWF) and the defect in blood coagulation (by raising the factor VIII level). Desmopressin (1-diamine-8-D-arginine vasopressin, DDAVP) is used to increase VWF/factor VIII release. Most experience in treating individuals with VWD is with intravenous infusion, with which the response is rapid (peak VWF levels within ~45–90 minutes of infusion). Doses may be repeated at intervals of 12 to 24 hours for continued bleeding or for postoperative use. Pediatric dosage is 0.3 µg/kg IV.

VWF/factor III infusion can be given for more severe cases. Plasma-derived factor VIII concentrates that contain VWF in high molecular weight can be used, and cryoprecipitate contains multimeric VWF. Humate-P (Centeon, L.L.C., King of Prussia, PA) and Alphanate (Grifols, S.A., Barcelona, Spain) are other forms of factor VIII/VWF. Aminocaproic acid (Amicar, Pfizer Pharmaceuticals, New York, NY) inhibits fibrinolysis via inhibition of plasminogen activator substances and, to a lesser degree, through antiplasmin activity. Amicar (pediatric dosing 100 mg/kg/dose orally every 4–6 hours) can be useful in cases of mucosal bleeding.

1.3 Obstructive Sleep Apnea

◆ Key Features

- Sleep apnea is a cessation of breathing during sleep.
- Sleep apnea may be central, obstructive, or mixed.
- Obstructive sleep apnea is caused by upper airway collapse and narrowing.
- Cardinal symptoms are disruptive snoring, witnessed apneas, and daytime sleepiness.

Obstructive sleep apnea (OSA) is characterized by repetitive episodes of complete (apnea) or partial (hypopnea) upper airway obstruction occurring

during sleep. Apnea is a cessation of breathing for at least 10 seconds. Hypopnea is a transient reduction of breathing for less than 10 seconds. Hypopneas must be associated with oxygen desaturation of at least 4%, whereas apneic events are not required to be associated with a decrease in oxygen saturation.

◆ Epidemiology

Estimates of the prevalence of sleep-disordered breathing vary widely in the literature, with most quoting at least 2 to 4% of the general population. Men have a higher prevalence than women (3–8 times), except that risk equalizes in postmenopausal females. The prevalence increases with age and body mass index (BMI) above 25, but it may remain undiagnosed in the majority of patients. Other risk factors include family history, obesity, left ventricular heart failure, advanced age, and allergy.

◆ Clinical

Signs and Symptoms

The signs and symptoms of sleep apnea include witnessed nocturnal apnea events, snoring, daytime sleepiness, headaches, depression, restless sleep, trouble concentrating, irritability, nighttime awakenings or gasping for air, and decreased libido.

Differential Diagnosis

Sleep disturbance and respiratory events can be caused by nonobstructive alveolar hypoventilation, asthma, chronic obstructive pulmonary disease, congestive heart failure, narcolepsy, periodic limb movement disorder, sleep deprivation, and medication, drug, and alcohol use. The differential diagnosis may also include laryngospasm related to gastroesophageal reflux disorder (GERD).

◆ Evaluation

History

The history taken from the patient and his or her bed partner will include reports of snoring, witnessed apneas or gasping events at night, daytime sleepiness, decreased libido or sexual dysfunction, and motor vehicle or work accidents. Bed partner reporting is helpful to evaluate a pretest risk of OSA but is not a good predictor of the severity of sleep-disordered breathing. The Epworth Sleepiness Scale (8 questions scored 0–3) is often used to assess daytime sleepiness; score >11 may correlate with OSA.

Physical Exam

- Body habitus (weight, BMI)
- Congenital craniofacial abnormalities

- Oral exam: large tongue base, elongated soft palate, enlarged uvula, narrow oropharyngeal inlet, modified Mallampati and tonsil staging (**Table 1.4**)
- Nasal exam: deviated septum, hypertrophic conchae, polyposis nasi
- Facial skeleton abnormalities: e.g., retrognathia, midface hypoplasia, craniofacial anomalies
- Neck circumference: men ≥ 17 inches (≥ 43 cm), women ≥ 15.5 inches (≥ 39 cm) is predictive of OSA

Imaging

Consider a chest X-ray to rule out right-sided heart failure.

Labs

Laboratory testing is not necessary.

Other Tests

Flexible fiberoptic nasopharyngoscopy can be useful to assess the retropalatal and retrolingual spaces for narrowing or dynamic collapse. Some surgeons utilize a Müller maneuver (patient inspires with closed mouth and nose) to assess for possible collapse that may indicate regions of anatomic concern during OSA events. Drug-induced sleep endoscopy (DISE) is a better predictor of possible anatomic sites of obstruction. This procedure involves flexible endoscopy with the patient under sedated anesthesia in an attempt to reproduce conditions similar to natural sleep.

Nocturnal *polysomnography* (PSG), or sleep study, is the gold standard for diagnosing and quantifying obstructive sleep apnea; multiple physiologic parameters are measured while the patient sleeps in a laboratory. Data are collected in the presence of a qualified technician. Typical parameters in a sleep study include eye movement observations (to detect rapid-eye-movement [REM] sleep), electroencephalography (EEG; to determine arousals from sleep), chest wall and abdominal monitors, nasal and oral airflow

Table 1.4 Mallampati and tonsil staging

Mallampati scoring		Tonsil grading	
Class I	Soft palate, uvula, faucial arches, pillars visible	0	Tonsils absent
Class II	Soft palate, portion of the uvula and faucial arch visible	1	Tonsils occupy $\leq 25\%$ of interpillar space
Class III	Soft palate and base of uvula visible	2	Tonsils occupy 26–50% of interpillar space
Class IV	Only hard palate visible	3	Tonsils occupy 51–75% of the interpillar space
		4	Tonsils occupy $\geq 75\%$ of the interpillar space

measurements, an electrocardiogram (ECG), an electromyogram (to look for limb movements that cause arousals), and oximetry.

Obstructive apnea is defined as an apneic event associated with increased respiratory effort throughout. *Central apnea* is an apneic event without associated respiratory effort. *Mixed apnea* begins as central, but then respiratory effort resumes.

The *apnea-hypopnea index (AHI)* is the most common summary measure used to describe respiratory disturbances during sleep. AHI is the total number of episodes of apnea and hypopnea during sleep, divided by the hours of sleep time. AHI values can be summarized for the sleep period or computed for different sleep stages and body position. An AHI >5 is required for the diagnosis of OSA; mild OSA is defined as AHI values of 5 to 15, moderate OSA for values of 16 to 30, and severe OSA for values ≥ 30 .

The *respiratory disturbance index (RDI)* is a similar measure to AHI and includes apneas, hypopneas, and respiratory-related arousal events (arousal with some respiratory perturbations that does not meet criteria for apnea or hypopnea).

The *arousal index (AI)* is the number of arousals per hour of sleep, which can be computed from an EEG. The AI may be correlated with AHI or RDI, but ~20% of apneas and desaturation episodes are not accompanied by arousals, or other causes of arousals are present. Some sleep centers utilize portable home monitors that measure only heart rate, pulse oximetry, and nasal airflow to diagnose OSA.

In *pediatric sleep studies*, EEG traces have different categories of activities. Apneic events are counted if duration is at least 2 baseline breaths. Hypopneas are associated with at least 3% O₂ desaturation or arousal.

◆ Treatment Options

Medical

Mild OSA may be managed conservatively by weight loss or avoidance of alcohol and certain medications. *Weight loss* is the simplest treatment for OSA in obese patients. Even a modest (10%) weight loss may eliminate apneic episodes by reducing the mass of the posterior airway. Positional mild OSA may be improved as well by sleep positioning maneuvers (e.g., avoidance of supine position by those with supine-predominant OSA).

Continuous positive airway pressure (CPAP) or *bilevel positive airway pressure (BiPAP)* treatment is the primary medical management for patients who have symptomatic mild or any moderate or severe obstructive sleep apnea. CPAP improves upper airway patency by the application of positive pressure to the collapsible upper airway. CPAP use can improve multiple cardiac and respiratory parameters as well as GERD, ED, and depression. Effective pressures typically range from 3 to 15 cm H₂O. The most common problem with CPAP is noncompliance. BiPAP independently provides inspiratory and expiratory pressures. CPAP failure may be caused by perceived discomfort, claustrophobia, nasal congestion, inadvertent dislodgement of mask during sleep, and panic attacks. *Contraindications* include cerebrospinal fluid (CSF) leak, pneumocephalus, pneumothorax or bullae, aspiration risk, and hypotension.

Oral appliances may be useful. *Mandibular advancement devices* are designed to advance the mandible forward and increase the retrolingual space or, at the very least, reduce mandible and tongue retrusion with sleep.

Surgical

A variety of procedures may improve OSA in properly selected patients. The challenge is identifying the likely anatomic site of obstruction to be corrected. Many surgeons advocate a potentially staged approach to treat additional sites if improvement is not achieved with the initial procedure. Sites of obstruction may include the nasal airway (septum, conchae, nasal valve), the palatine tonsils and soft palate/uvula, the base of tongue/lingual tonsils, and the mandible. Common procedures include:

- Septoplasty with or without inferior concha reduction may improve nasal breathing and reduce mouthbreathing. It may enable CPAP to be used at a lower pressure with a nasal-only device, but it is unable to generate any significant change in AHI.
- Adenoidectomy or adenotonsillectomy may be performed, especially in children with OSA.
- Uvulopalatopharyngoplasty (UPPP) involves the removal of part of the soft palate, some or all of the uvula and redundant oropharyngeal tissues, including the tonsils, and then subsequent resuspension of pharyngopalatal tissues, with various described techniques (palatal advancement or uvulopalatal flaps).
- Tongue base reduction procedures include a midline glossectomy, lingual tonsillectomy (with or without transoral robotic technique), radiofrequency/Coblation (Smith & Nephew, London, UK) treatment of base of tongue. Risks: bleeding, hematoma, lingual artery or nerve injuries, dysgeusia.
- Orthognathic surgery may consist of genioglossal, mandibular, or maxillomandibular advancement. Risks include nonunion, deformity, temporomandibular joint (TMJ) disorder, and nerve injury.
- Base of tongue or hyoid suspension may be attempted.
- Hypoglossal nerve stimulation implant involves a sensing lead from the intercostal muscles, a battery-powered implantable pulse generator, and a stimulation lead secured to a hypoglossal nerve. Excellent outcomes have been reported.
- Tracheotomy is the most effective therapeutic maneuver for obstructive apnea, as it bypasses the obstructed region, but it is a procedure of last resort. It is indicated for patients who are most severely affected by OSA or obstructive sleep hypopnea and who cannot utilize CPAP effectively.

◆ Outcome and Follow-Up

Prognosis for improvement is excellent with proper treatment. Polysomnography is often repeated ~3 months following surgical treatment or following weight loss achievement. Surgical success is often defined as at least a 50% reduction in AHI or reduction of AHI to below 5. A 50% reduction may not

be a cure, and either retitration of CPAP or additional staged treatment may be required.

OSA is frequently undiagnosed and untreated and can have long-term sequelae, including poorly controlled hypertension, arrhythmias, cerebral vascular accidents, pulmonary vascular disease, heart failure, and further weight gain with its associated morbidities. The adverse effects of hypersomnolence may include loss of employment, motor vehicle accidents, and sexual dysfunction.

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